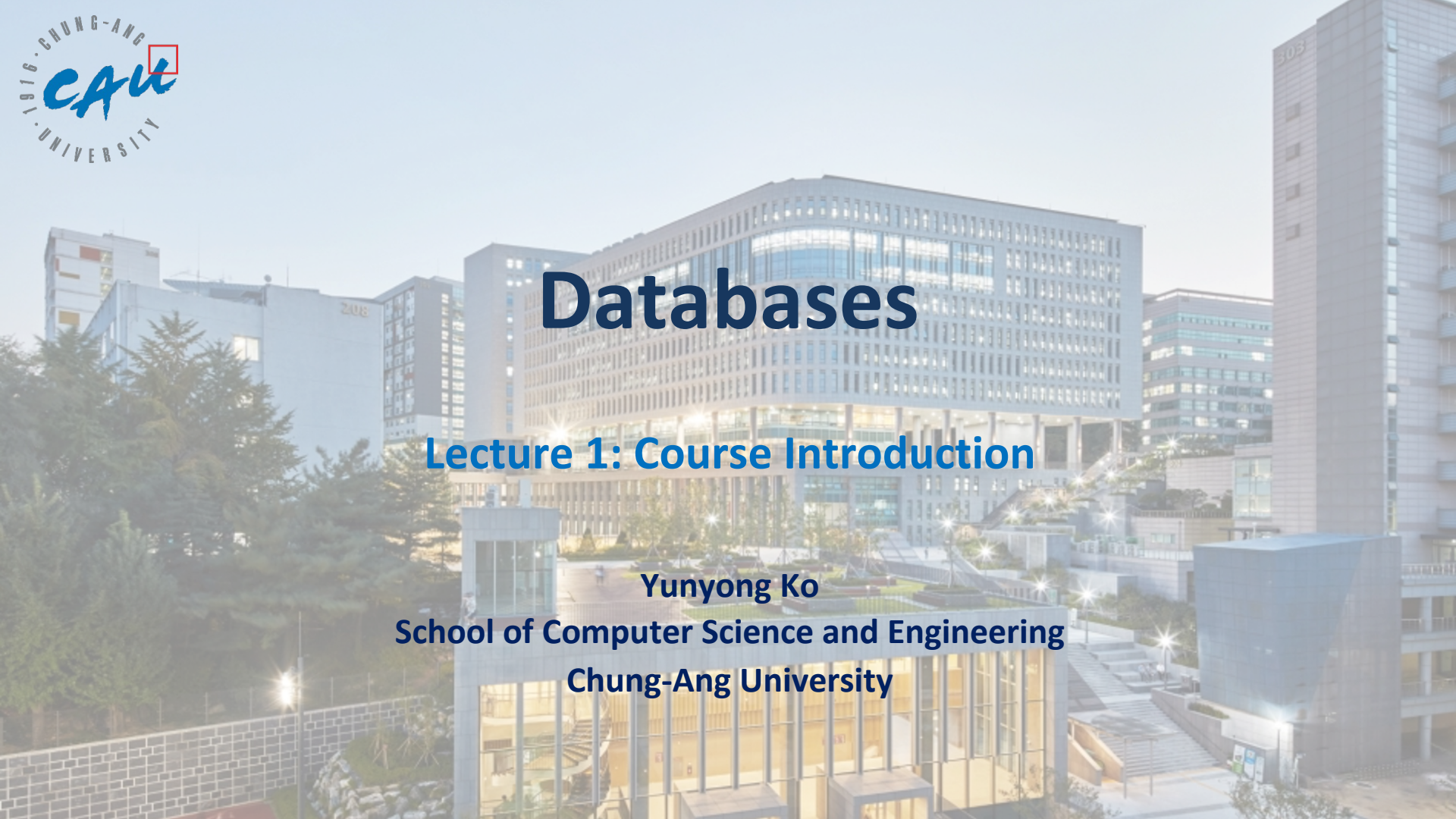


Databases

Lecture 1: Course Introduction

Yunyong Ko
School of Computer Science and Engineering
Chung-Ang University

Outline

❑ Course Overview

- Course and Instructor information
- Lecture materials and grading policy
- Course goals: What we will learn



❑ Introduction to Database Systems

- Definition and applications
- Characteristics and advantages

Course Overview

- **Course Title: Databases (59190)**
- **Lecture Hours**
  - #1: Tuesday 11:00 am - 12:50 pm and Friday 11:00 am - 11:50 am (310-B210)
  - #2: Tuesday 1:00 pm - 2:50 pm (310-729) and Friday 1:00 pm - 1:50 pm (310-728)
- **Course Homepage: e-class**
  - All materials will be uploaded on the e-class before every class
- **Language: English**
  - I can give a supplementary explanation in Korean

Instructor and Teaching Assistants

❑ Instructor: Yunyong Ko (고윤용), Assistant Professor @ CAU

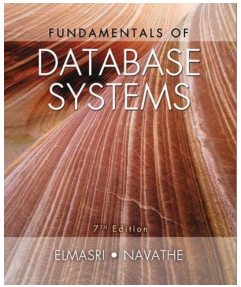
- Research interests: Machine Learning (ML) and Data Science (DS)
  - Lab: **M**achine **IN**telligence and **D**ata **S**cience (**MINDS**) Lab.
- E-mail: yyko@cau.ac.kr
- Office: Building 310, Room 744
- Homepage: https://yy-ko.github.io (Lab: https://www.minds.cau.ac.kr/)

❑ TA Information

- Joohee Cho (조주희), joohee2008@cau.ac.kr
- Sooho Moon (문수호), moonwalk725@cau.ac.kr
- Office: Building 208, Room 516
- Office Hours: 3:00 pm – 3:50 pm (Tuesday)

Lecture Materials

❑ Textbook



- "***Fundamentals of Database Systems***" (7th edition)
  - Authors: Ramez Elmasri and Shamkant B. Navathe

❑ Slides + Hands-on Practices

- Lecture slides (pdf) per each week and additional materials (hands-on practices)
- All materials will be uploaded on the e-class before every class

❑ Prerequisites: Data Structure, Basic Programming

- This is **NOT** a programming class
- This course includes some programming assignments

Grading Policy

- **Exams: Midterm (40%) + Final (40%)**
  - Closed-book test (Textbook and lecture slides)
  - **NO** tolerance on cheating (Cheating will result in **F** grade)

A+ or **A0** <= 45%
B+ or **B0** <= 80%
C+ or **C0** > 80%

This cut-off can be adjusted slightly

- **Assignments (15%): 2 Assignments**
  - **Submission due**: **11:59 PM** on the due date
  - **Late submission**: +1 days (-25%), +2 days (-50%), +3 days (No score)
    - Contact the TA if you have any difficulties in submitting the assignment
  - **Plagiarism**: **NO** score if plagiarism is detected
    - Plagiarism detection tools will be used to check your code
  - **Programming Languages**: Python
- **Attendance (5%)**
  - **No** grade if you are absent for more than 30% of all the classes

Things to Keep in Mind

❑ Frequently Check the e-class

- All announcements and lecture materials will be uploaded on the e-class
  - Assignments, examinations, class cancellation and makeup class, etc.

❑ Q&A

- Contact me and TA if you have any questions

❑ Academic Integrity Policy

- **NO** tolerance on any violation (cheating, plagiarism, …)
  - Feel free to discuss with other members, but write down your own solution **independently**

❑ Etiquette

- **Non-hostile environment**: be respectful to one another
- **Mobile device**: put your phone/tablet on silent/vibrate mode
- **Email**: write emails properly (to TAs or to Professor)

What We Will Learn

❑ Let's Learn *How Database Systems Work*

- Part 1: How to "Use" Database Systems
  - ❑ We learn the basic concepts and key functions of DBMS with hands-on practices
- Part 2: How to "Implement" Database Systems
  - ❑ We learn the internal architecture of DBMS
    - When something gets wrong, we need a DBA (DB administrator)
    - *AI will not be able to fully replace DBAs in the near future*





Knowing How It Works == Using + Fixing

What We Will Learn: Part 1

❑ Part 1: How to “Use” Databases (Weeks 1 - 10)

- Introduction to Database Systems
- Conceptual Data Modeling: ER model
- Relational Database
- Basic / Advanced SQL
- Relational Algebra
- Functional Dependencies and Normalization
- AI for Database Systems: NL2SQL



What We Will Learn: Part 2

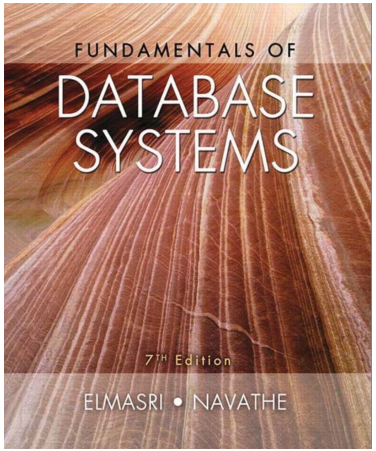
❑ Part 2: How to “Implement” Databases (Week 11-16)

- File Structures and Storage Management
- Indexing and Physical Database Design
- Query Processing and Optimization
- Transaction and Concurrency Control
- Database Recovery Techniques



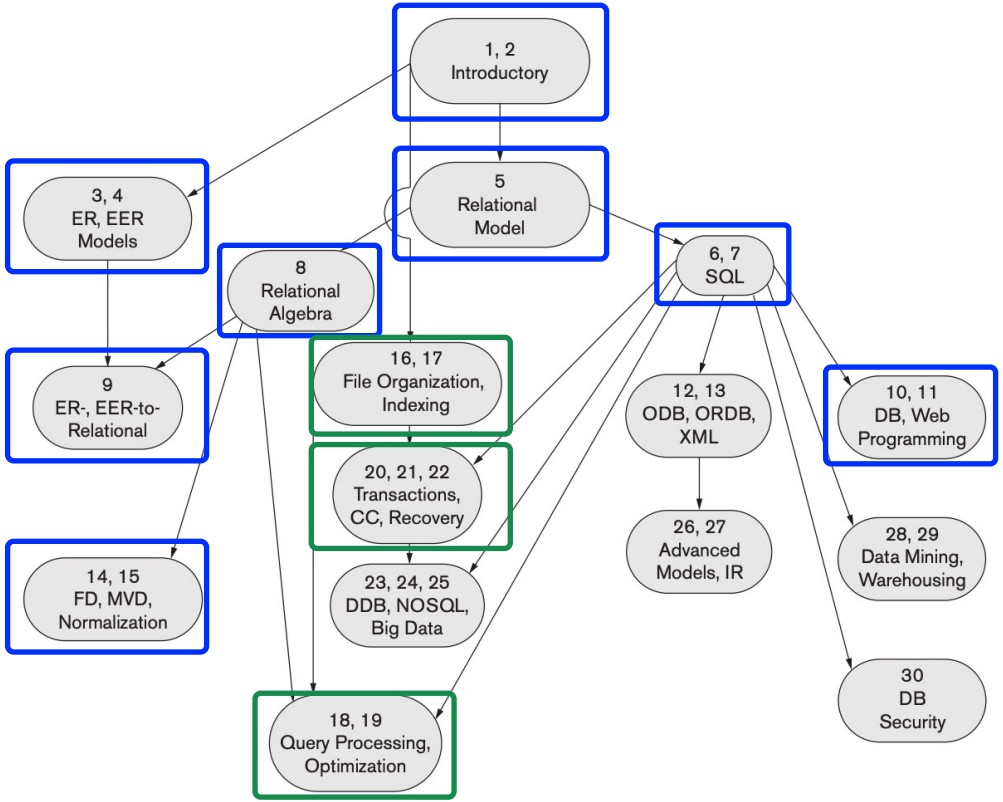
What We Will Learn (Cont.)

❑ Textbook View



Part 1 | Part 2

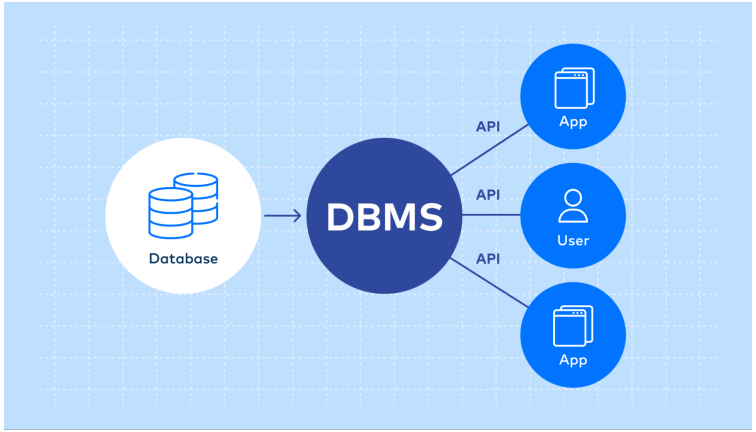
13 chapters | 7 chapters

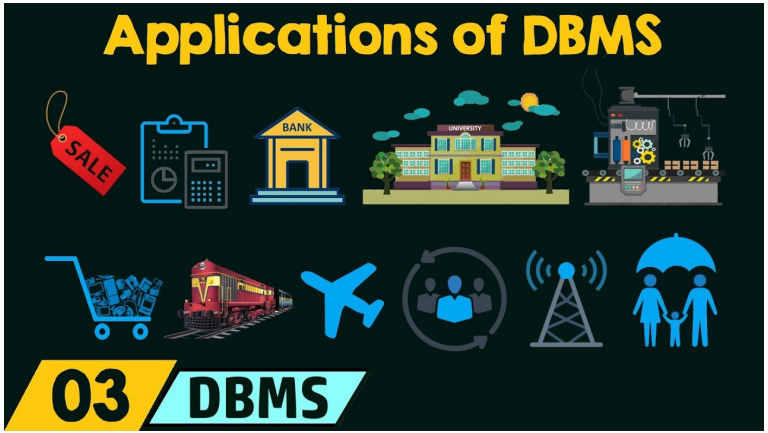


Course Goals

❑ By the End of This Course, You Will be Equipped with:

- The knowledge to comprehend *the principles of databases and DBMS*
- The ability to fluently use *basic/advanced SQL with AI*
- The skills to implement *database systems for real-world applications*





Course Schedule (Tentative)

Week		Topic	Week		Topic
1	2026.09.01	**Course Introduction**	9	2026.10.27	AI for Databases: NL2SQL
	2026.09.04	No class		2026.10.30	Hands-on Practice
2	2026.09.08	ER Model (Video Lecture)	10	2026.11.03	AI for Databases: NL2SQL
	2026.09.11	Hands-on Practice		2026.11.06	Hands-on Practice
3	2026.09.15	Relational Database (Video Lecture)	11	2026.11.10	Disk Storage and File Structure
	2026.09.18	Hands-on Practice (Video Lecture)		2026.11.13	Parallelizing Disk Access
4	2026.09.22	Basic SQL (Video Lecture)	12	2026.11.17	Indexing for Physical Database Design (Assignment #2)
	2026.09.25	No class (추석 연휴)		2026.11.20	Advanced Indexing Techniques
5	2026.09.29	Advanced SQL (Assignment #1)	13	2026.11.24	Query Processing
	2026.10.02	Hands-on Practice		2026.11.27	Query Optimization
6	2026.10.06	Relational Algebra	14	2026.12.01	Transaction Processing
	2026.10.09	Hands-on Practice (Video Lecture, 한글날)		2026.12.04	Concurrency Control and Recovery
7	2026.10.13	Functional Dependencies and Normalization	15	2026.12.08	Advanced Topic: TBD
	2026.10.16	Hands-on Practice		2026.12.11	No class (Q&A)
8	2026.10.23	**Midterm Exam**	16	2026.12.18	**Final Exam**

Outline

- Course Overview
  - Course and Instructor information
  - Lecture materials and grading policy
  - Course goals: What we will learn

- **Introduction to Database Systems**
  - Definition and applications
  - Characteristics of DB systems



What is a Database System?

❑ Database (DB)

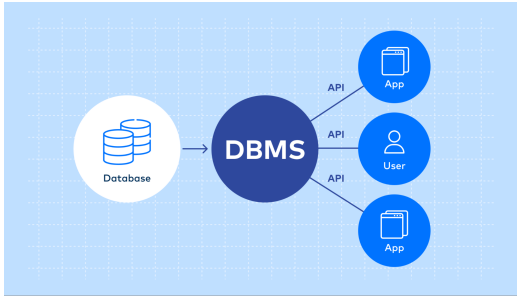
- A logically coherent collection of inter-related data
  - ❑ Managed by a ***Database Management System***

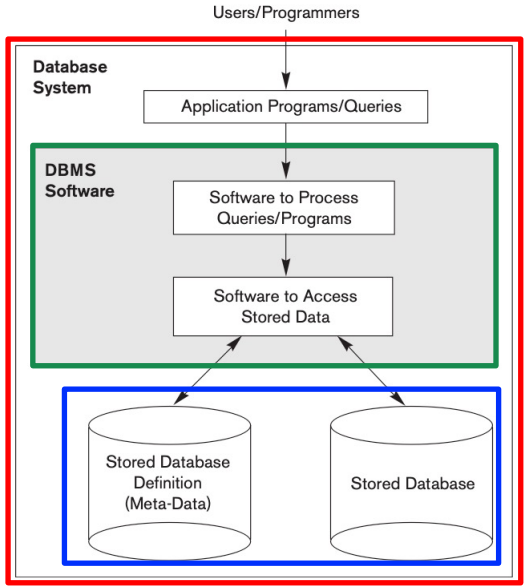
❑ Database Management System (DBMS)

- A software that is designed to efficiently "**manage**" data in a structured way
  - ❑ ***Defining***, ***constructing***, ***manipulating***, and ***sharing*** databases
  - ❑ e.g., IBM DB2, Oracle, MS SQL Server, MySQL, SQLite, …

❑ Database System

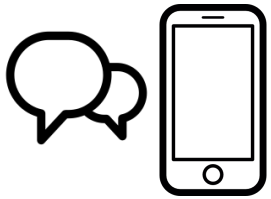
- Databases + DBMS software





Why Use Database Systems?

❑ Simple Real-world Example: Message App



- We want to manage data for a simple message application
- Necessary data
  - ❑ Information of Friends: Name, Phone number, isAdvertisement, …
  - ❑ Information of Messages: Message texts, sender information, date, …

❑ Using Simple Text Files

- Store each line in the file
- Separate each attribute by a comma

Friends (name, number, ad)

"Jonghyeok" , 010-1234-1234 , No

"McDonald" , NULL , Yes

"Starbucks" , NULL , Yes

Message (content, sender, date)

"Class.." , "Jonghyeok", 2409020130

"Today …" , "McDonald" , 2409020900

"Get promo.." , "Starbucks" , 2409020900

Why Use Database Systems? (Cont.)

❑ A Number of Real-world Applications/Services

- E-Commerce websites
  - ❑ Customers, items, transactions, shipping, …
- Banking and finance sector
  - ❑ Users, account details, financial transactions, …
- Social media sites
  - ❑ Users, contents, social information, messages, …
- University/Library management systems
  - ❑ Students, courses, books, authors, …
- Airline/railway reservation systems
  - ❑ Customers, ticket appointments, rail routes, …
- …





Why Use Database Systems? (Cont'd)

❑ Data Integrity

- How can we manage the total length of message?
- What if a user adds duplicate friend information by mistake?
- What if there are friends with the same name?



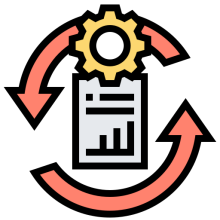
❑ Data Management

- How can we find a particular message quickly?
- What if multiple users try to read/write at the same time?



❑ Data Durability

- What if the machine crashes while the app is writing new entries?
- How to restore files after crash?



There are a number of technical issues to be addressed😭
Let's use DBMS!

Characteristics of Database Systems

1) **Self-Describing Nature**

2) **Data Abstraction**

3) **Support of Multiple Views of the Data**

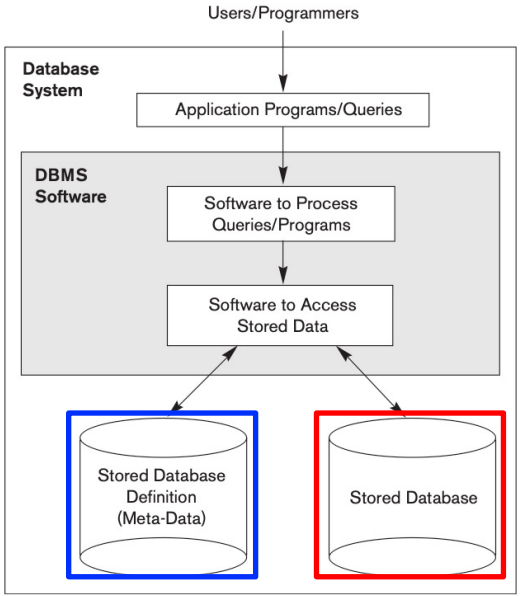
4) **Sharing of Data and Multiuser Transaction Processing**

1) Self-Describing Nature

❑ A Database System Contains the DB Itself and Meta-Data

- **Meta-data**: *a definition of the database structure and constraints*
  - ❑ Meta-data is stored in the DBMS catalog
  - ❑ It is necessary to maintain the DB

Q. How can we define them? → SQL in Weeks 4-5



RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

An example of a database catalog

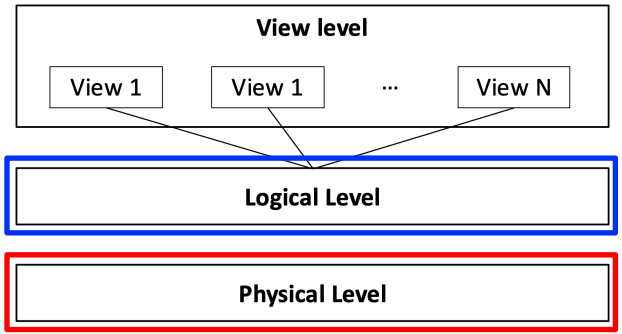
COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
….	….	…..
….	….	…..
….	….	…..
Prerequisite_number	XXXXNNNN	PREREQUISITE

2) Data Abstraction

❑ Insulation between Programs and Data

- A database system provides a ***conceptual representation (logical level)*** of data
  - ❑ e.g., objects, their properties, and their interrelationship
- It does not include the complex structures of a database (***physical level***)
  - ❑ e.g., how the data is stored or how the operations are implemented
- *A change of data structure does not affect programs* (**program-data independence**)



Data Item Name	Starting Position in Record	Length in Characters (bytes)
Name	1	30
Student_number	31	4
Class	35	1
Major	36	4

File storage format for a STUDENT record



3) Support of Multiple Views

❑ Different Users May Require Different Views of the Data

- View: **virtual data** derived from the database, but not explicitly stored
  - ❑ i.e., some users may not need to be aware of whether the data is stored or derived

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

University Database

TRANSCRIPT

Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

View 1: accessing/printing the transcript of each student

COURSE_PREREQUISITES

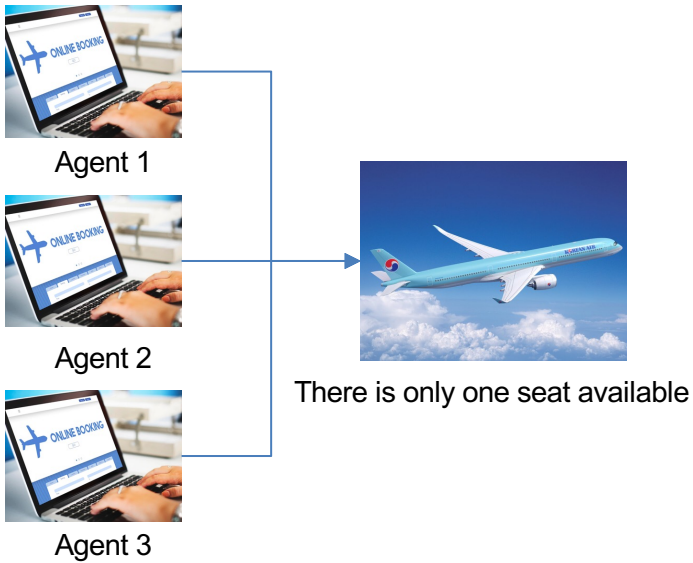
Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

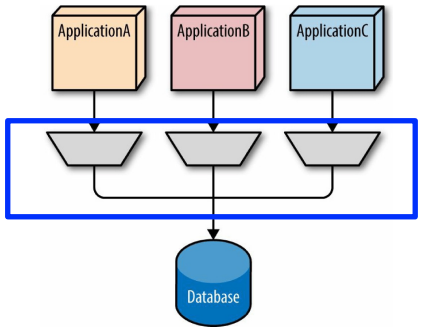
View 2: checking the prerequisites of each course

4) Sharing of Data & Multiuser Processing

❑ Multiple Users to Access the Database Simultaneously

- Data for multiple Apps can be integrated/maintained in a single database
  - ❑ e.g., several reservation agents try to assign a seat on an airline flight
- **Concurrency control** is needed to ensure multiuser *transactions*
  - ❑ **Transaction**: a collection of operations that perform a single logical function



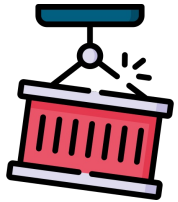


Transactions should hold the **ACID** properties
Atomicity, **C**onsistency, **I**solation, **D**urability
→ Concurrency control in Week 14

When Not to Use a DBMS?

❑ Overheads of DBMS

- High initial investment in hardware, software, and training
- The generality that a DBMS provides for defining and processing data
- Security, concurrency control, recovery, and integrity functions



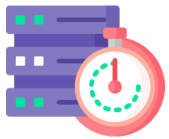
Too heavy..😭

❑ More Desirable to Use Customized Databases

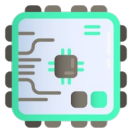
- Simple and well-defined applications that are *not expected to change at all*
- *Strict real-time requirements* that may not be met
- *Embedded systems with limited storage capacity* (no need general-purpose DBMS)
- *No multiple-user* access to data

No Addition
No Deletion
No Update





Real-time Requirements



Limited Capacity

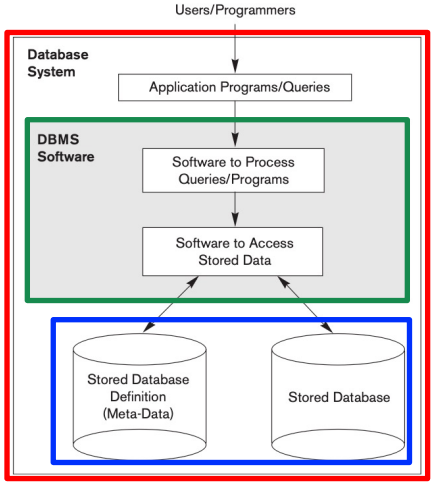


No Multiple-user Access

Takeaways: Database System Concepts

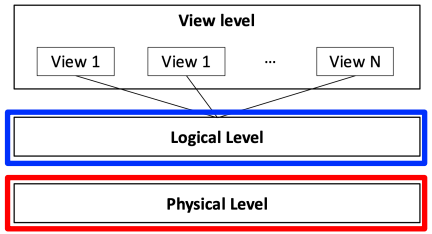
❑ Introduction to Database

- Database, DBMS, and Database System



❑ Characteristics of DB Systems

- Self-describing nature
- Data abstraction
- Support of multiple views of the data
- Sharing of data and multiuser transaction processing



TRANSCRIPT

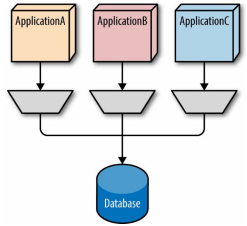
Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

View 1

COURSE_PREREQUISITES

Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

View 2



Q&A:
Course Introduction